

# Feature Engineering

2026.04.21

Data sets  
in tutorials



Data sets in  
the wild

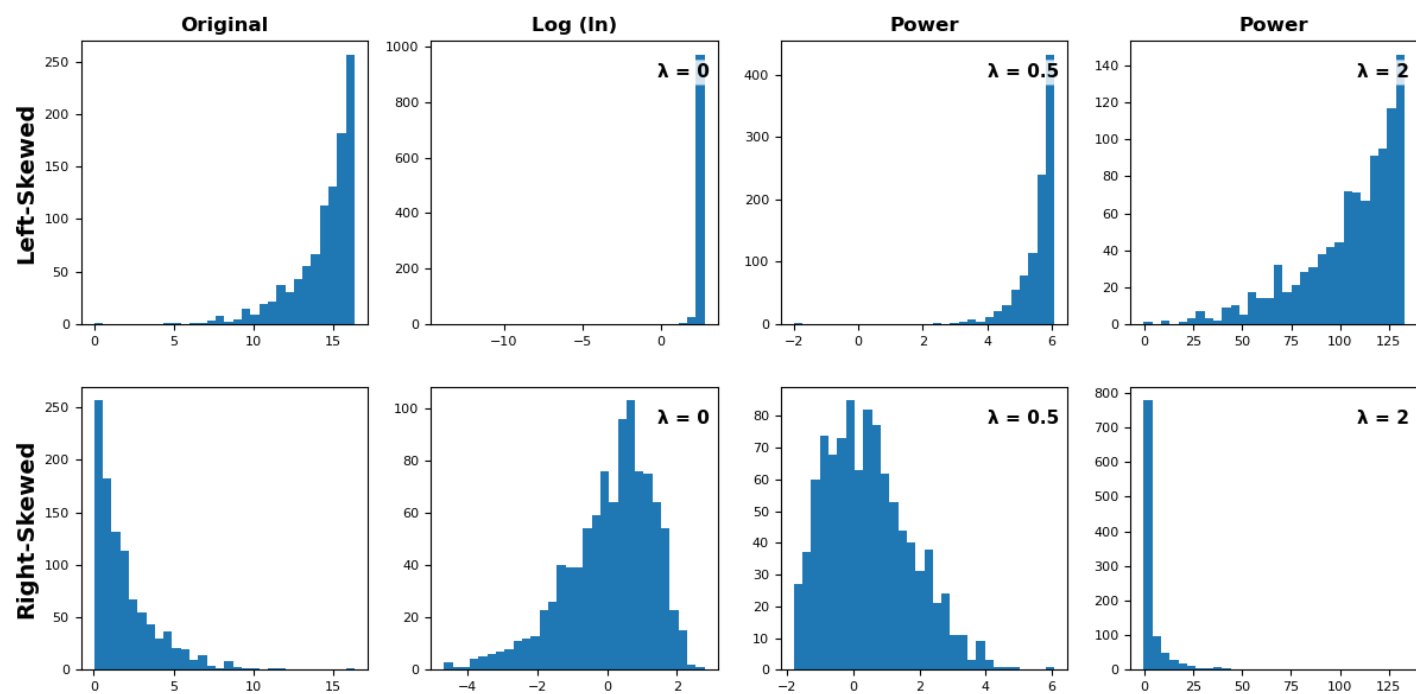


Feature engineering: how we  
represent the data

# Preprocessing

- Fix distributions (skew, outliers)
- Impute missing values
- Encoding categorical features
- Scale (as needed in model building pipeline)

# Transformations for distribution/skew

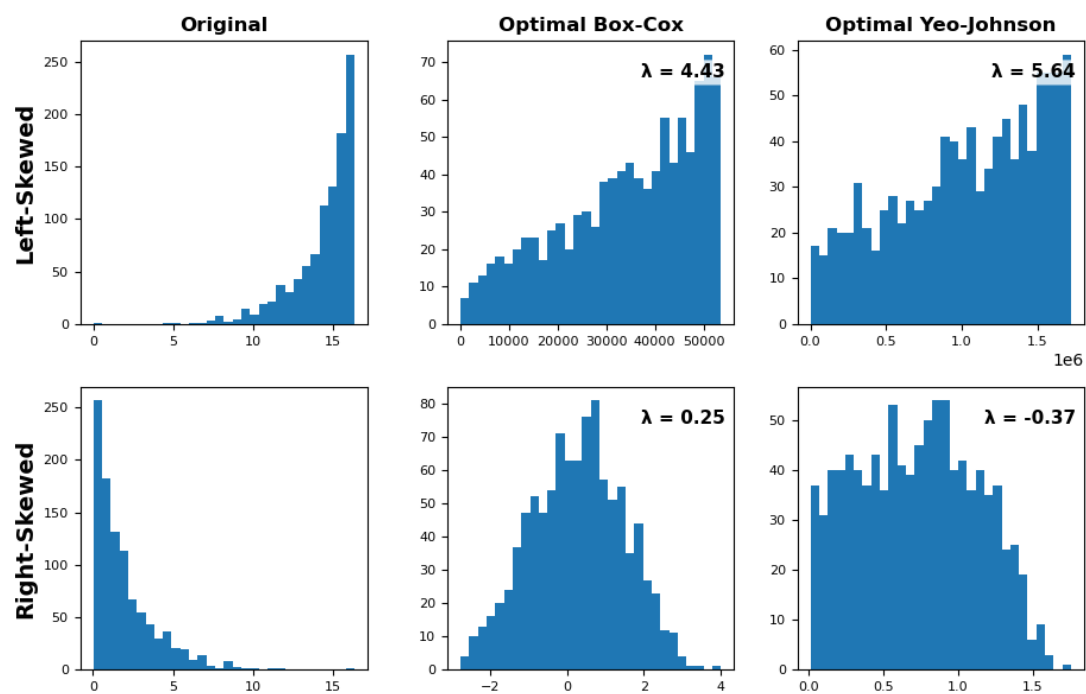


\* More important for linear models and neural networks than for tree-based models

# Box Cox

$$y(\lambda) = \begin{cases} \frac{x^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0 \\ \ln(x) & \text{if } \lambda = 0 \end{cases}$$

# Box Cox and Yeo Johnson



# Box Cox/Yeo Johnson to simple transformations

| $\lambda$ range | Meaning                 | Simple approximation              |
|-----------------|-------------------------|-----------------------------------|
| $\approx 1$     | no change               | leave it                          |
| 0.5–1           | mild compression        | $\sqrt{x}$                        |
| $\approx 0$     | strong compression      | $\log(x)$                         |
| -1 to 0         | very strong compression | $1/x$                             |
| $< -1$          | extreme compression     | reconsider / use Box–Cox directly |
| $> 1$           | expansion               | $x^2$                             |

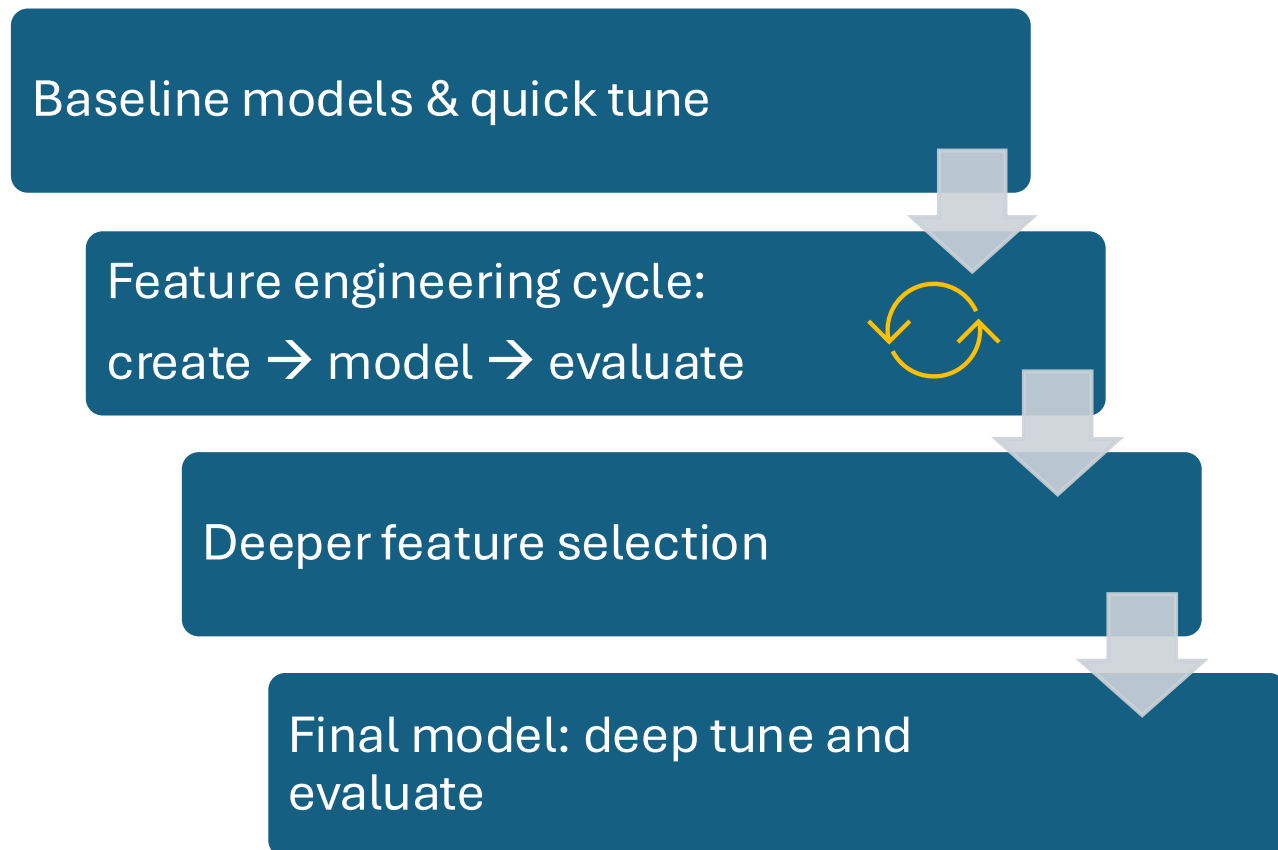
# When to transform?



# Encoding categorical features

- Consider combining rare categories
- One hot encoding
- Ordinal encoding
- Target encoding

# Feature engineering workflow



# Feature engineering: start here

- Interactions/ratios (hereditary principle)
- Clustering

# Feature-engine

- Structured preprocessing & encoding

# Stacking/ensembling

- Different models make different types of errors
- Once have features and have tuned, try combining models

# Stacking

- Averaging of predictions from multiple models (or weighted average)
- Meta-learner created using outputs from

# Wrap up

- Feature engineering creates options
- Modeling and evaluation decides what matters
- Most engineered features don't improve model performance